INCH-POUND

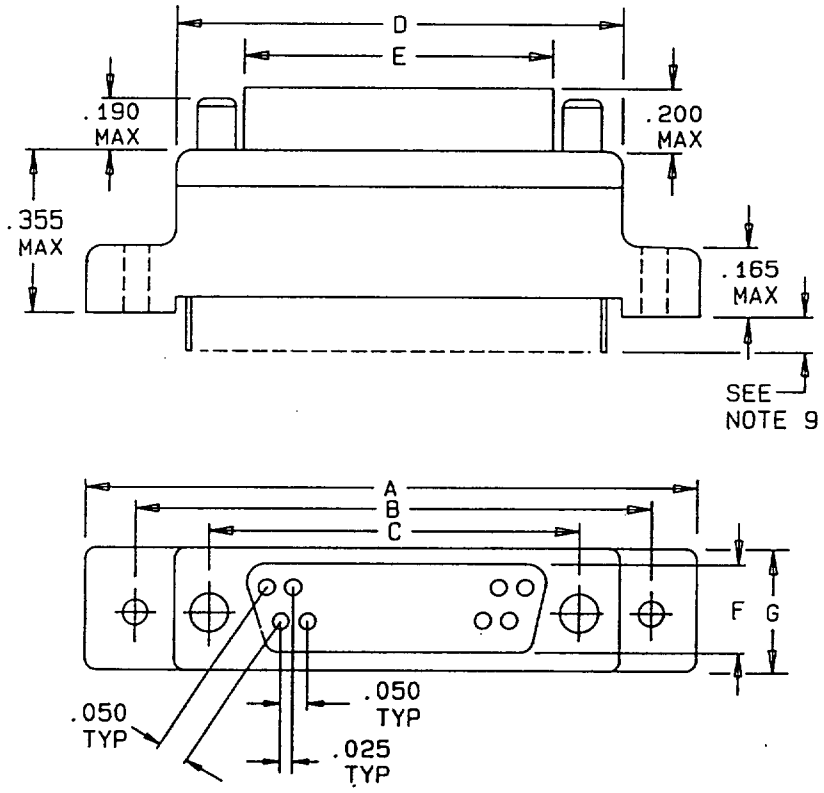
MIL-C-83513/22
11 Feb 1992

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, PLUG, MICROMINIATURE, POLARIZED SHELL, STRAIGHT, PIN CONTACTS, 2 ROW ,SOLDER TYPE, 9 THROUGH 37 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification Listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83513.



Inch	mm
.015	0.381
.025	0.635
.050	1.27
.109	2.775
.200	5.08
.355	9.017

FIGURE 1. Connector, plug, .050 spacing.

AMSC N/A FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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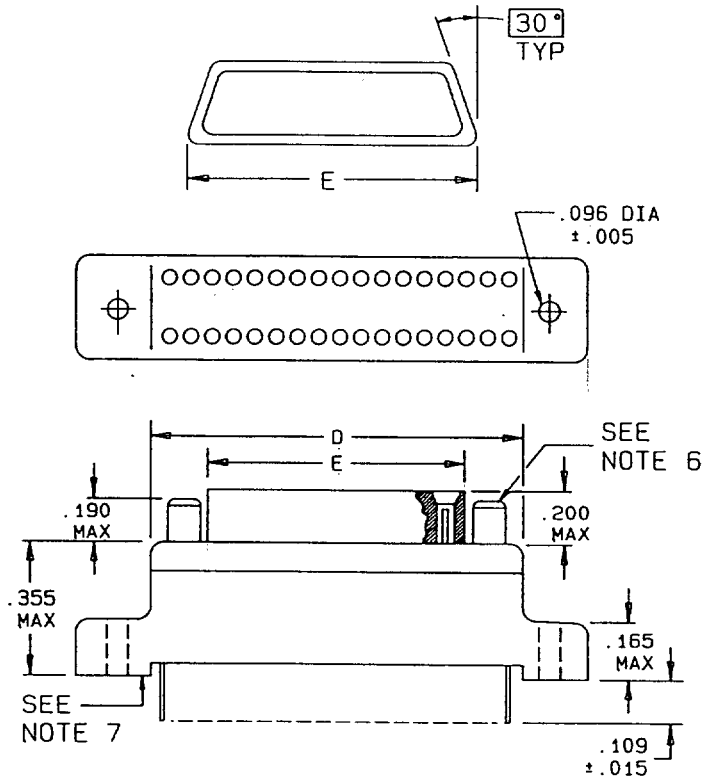


FIGURE 1. Connector, plug, .050 spacing. - Continued.

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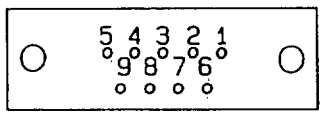
Number of contacts	A Max	B ±.007 0.18 mm	C ±.005 0.13 mm	D Min	E Max	F Max	G Max
9	1.390 (35.31)	1.150 (29.21)	.565 (14.35)	.885 (22.48)	.334 (8.48)	.185 (4.70)	.305 (7.75)
15	1.390 (39.12)	1.150 (33.02)	.715 (18.16)	.945 (24.00)	.484 (12.29)	.185 (4.70)	.305 (7.75)
21	1.690 (40.64)	1.450 (36.83)	.865 (21.97)	1.185 (30.10)	.634 (16.10)	.185 (4.70)	.305 (7.75)
25	1.740 (44.20)	1.500 (38.10)	.965 (24.51)	1.275 (32.39)	.734 (18.64)	.185 (4.70)	.305 (7.75)
31	2.040 (51.82)	1.800 (45.72)	1.115 (28.32)	1.575 (40.01)	.884 (22.45)	.185 (4.70)	.305 (7.75)
37	2.340 (59.44)	2.100 (53.34)	1.265 (32.13)	1.875 (47.63)	1.034 (26.26)	.185 (4.70)	.305 (7.75)

NOTES:

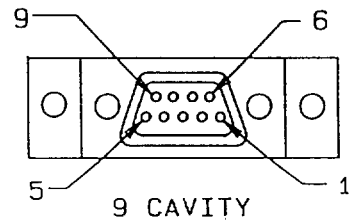
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. Termination organization cavity to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513.
5. Metal shell shall be of material in accordance with MIL-C-83513 for for class M parts.
6. Jackpost (permanently attached): Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35.
7. Plastic body shall conform to the requirements of type GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
8. Wire termination pins shall conform to QQ-W-343, no. 24 AWG copper.
9. Termination lengths available: .109 (2.77 mm), .140 (3.56 mm), or .172 (4.37 mm). The tolerance shall be ±.015 for all terminations lengths.

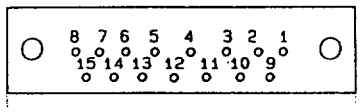
FIGURE 1. Connector, plug, .050 spacing. - Continued

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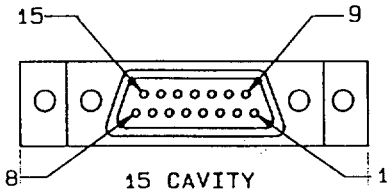


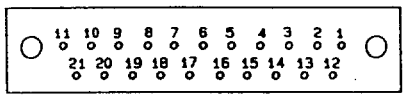
TERMINATION VIEW - STRAIGHT THROUGH



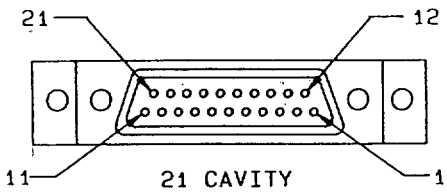


TERMINATION VIEW - STRAIGHT THROUGH





TERMINATION VIEW - STRAIGHT THROUGH

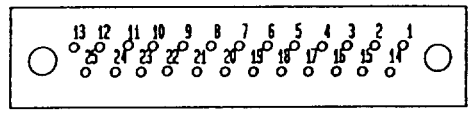


NOTE:

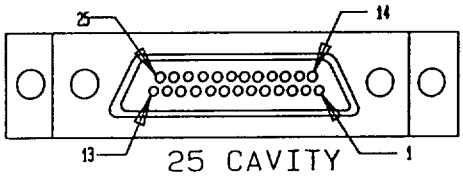
Engaging face of pin insert shown, cavity identification numbers are for reference only and do not appear on the part.

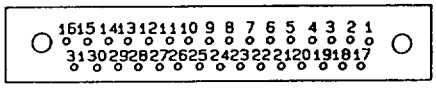
FIGURE 2. Insert arrangement.

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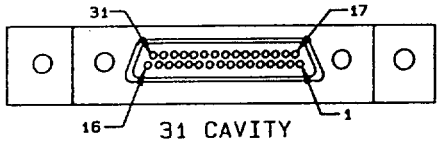


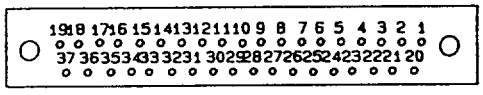
TERMINATION VIEW - STRAIGHT THROUGH



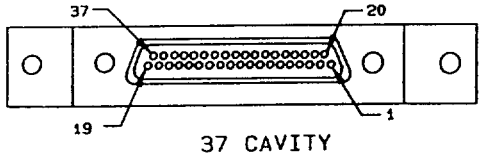


TERMINATION VIEW - STRAIGHT THROUGH





TERMINATION VIEW - STRAIGHT THROUGH



NOTE:

Engaging face of pin insert shown, cavity identification numbers are for reference only and do not appear on the part.

FIGURE 2. Insert arrangement. - Continued.

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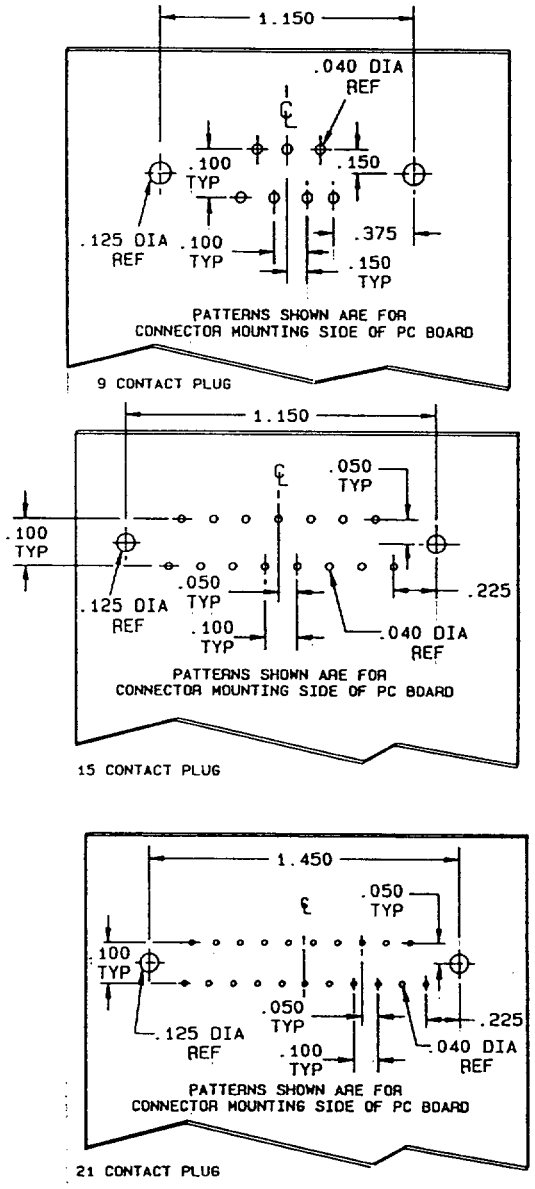


FIGURE 3. Layout arrangement.

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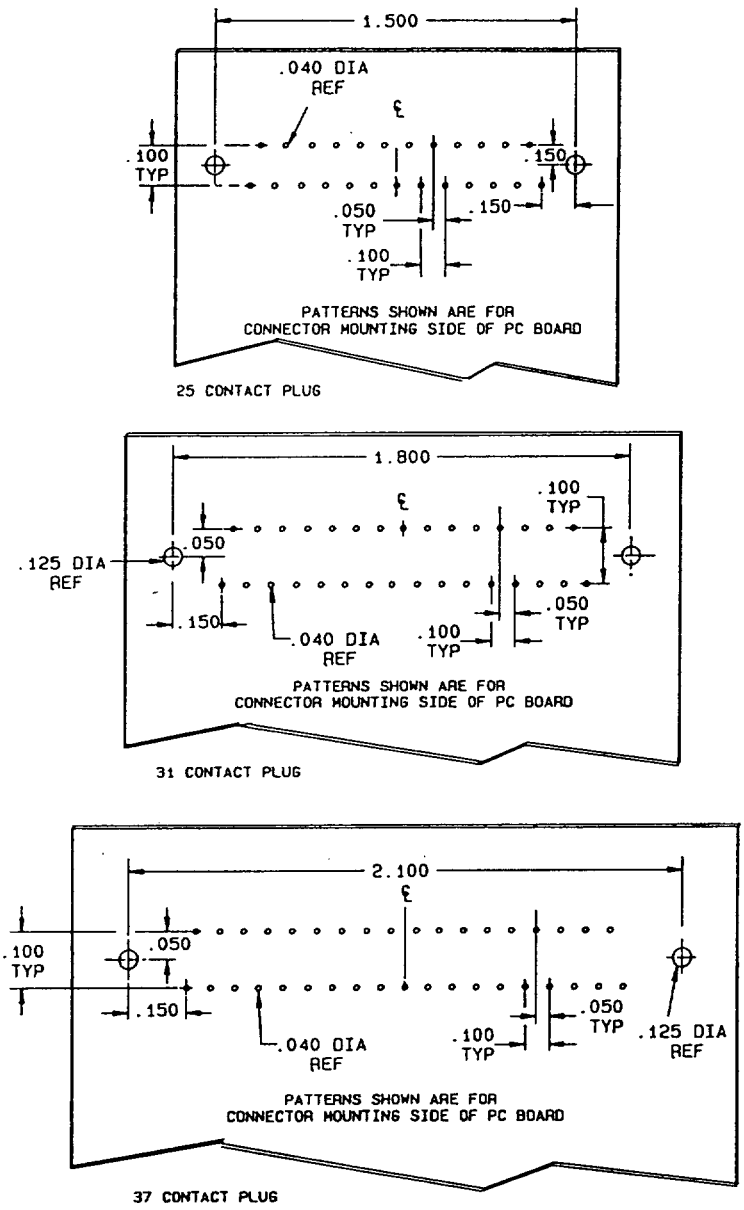


FIGURE 3. Layout arrangement. - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

Current rating, maximum: 3 amperes per contact.

Materials: The requirments for shell materials shall be in accordance with MIL-C-83513.

Mating plug: Shall conform to MIL-C-83513/2 and MIL-C-83513/4.

Plating of termination leads: Solder dipping of termination leads will be accomplished in SN60 or SN63 in accordance with QQ-S-571.

Part identification number: The PIN shall consists of the letter M, the basic number of the specification sheet, a letter from the insert, a numerical code for the termination length, and a letter code for the shell finish and hardware column.

M83513/22-	D	01	C	P
Specification sheet number	Insert arrangements (see figure 2) A=09 B=15 C=21 D=25 E=31 F=37	Termination length 01=.109 02=.140 03-.172	Shell finish C= cadmium N=electroless nickel (space application)	Hardware N=no jackpost P=jackpost attached T=threaded insert

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, SH
Air Force - 11, 17, 99

User activites:
Army - AT, ME
Navy - CG, MC

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project: 5935-3739-13)

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